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SHORTED LOAD DETECTION USING PWM LOOP

Abstract

A system for detecting short circuits including a driver having a first output and a second output, the driver configured to provide a first voltage to the first output and a second voltage to the second output; a loop filter coupled to the first rail at a first connection and to the second rail at a second connection; a first digital-to-analog converter (DAC) having a first DAC output coupled to the first connection and configured to provide a first DAC current to the first connection, and a second DAC output coupled to the second connection and configured to provide a second DAC current to the second connection; an assist DAC having a first assist output coupled to the loop filter and configured to provide a first assist current and having a second assist output coupled to the loop filter and configured to provide a second assist current; and a controller.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application 63/559,314, titled “Shorted Load Detection using PWM Loop,” filed on Feb. 29, 2024, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

1. Field of the Disclosure

[0002] At least one example in accordance with the present disclosure relates generally to load and short detection in circuits.

2. Discussion of Related Art

[0003] Amplifiers in devices may be damaged or have their operation impeded by various fault conditions. Some devices may protect amplifiers from fault conditions by shutting the amplifiers off.

SUMMARY

[0004] According to at least one aspect of the present disclosure a system for detecting short circuits is provided the system comprising: a driver having a first output and a second output, the driver configured to provide a first voltage and/or current to the first output and a second voltage and/or current to the second output; a loop filter coupled to the first rail at a first connection and to the second rail at a second connection; a first digital-to-analog converter (DAC) having a first DAC output coupled to the first connection and configured to provide a first DAC voltage and/or current to the first connection, and a second DAC output coupled to the second connection and configured to provide a second DAC voltage and/or current to the second connection; an assist DAC having a first assist output coupled to the loop filter and configured to provide a first assist voltage and/or current and having a second assist output coupled to the loop filter and configured to provide a second assist voltage and/or current; and a controller.

[0005] In some examples, the controller is further configured to adjust the output of the first DAC, the second DAC, based on an on resistance and an off resistance. In some examples the controller is configured to determine a value of the off resistance by: determining a first value based on a differential current and a stored value of the off resistance, the differential current being based on currents associated with the first rail and the second rail; and determining a second value based on a common mode voltage and a common mode current, the common mode voltage and the common mode current based on voltages and currents associated with the first rail and the second rail. In some examples, the controller is configured to increase the value of the off resistance responsive to determining that the second value is greater than the first value. In some examples, the controller is configured to decrease the value of the off resistance responsive to determining that the second value is less than the first value. In some examples, the second value is further based on a scaled value of a control voltage and/or current used to control a pulse-width-modulation of the driver. In some examples, the second value is equal to the common mode voltage less the common mode current and plus the scaled value of the control voltage. In some examples, the second value is equal to the common mode voltage less the common mode current. In some examples the first value is equal to a product of the differential current and the stored value of the off resistance. In some examples the controller is configured to determine a value of the on resistance by: determining a first value based on a 100% duty cycle pulse width modulation and an audio drive value; and determining a second value based on the 100% duty cycle pulse width modulation, a non-overlap factor, an ideal duty cycle, and a pulse drive value. In some examples the on resistance is incremented responsive to determining that the second value is greater than the first value. In some examples the on resistance is decremented responsive to determining that the second value is

less than the first value. In some examples the first value is further determined based on the non-overlap factor and the ideal duty cycle. In some examples the first value is product of the audio drive and first compositive value, wherein the first composite value is the 100% duty cycle pulse width modulation minus a second composite value, wherein the second composite value is a sum of the ideal duty cycle less a product of the 100% duty cycle pulse width modulation and the non-overlap factor. In some examples the first value of a product of the 100% duty cycle pulse width modulation and the audio drive. In some examples the second value is the pulse drive value times an adjusted drive value, wherein the adjusted drive value is the ideal duty cycle less the product of the 100% duty cycle pulse width modulation and the non-overlap factor.

[0006] According to at least one aspect of the present disclosure, a system is provided having a controller configured to perform all the operations of the controller discussed above; memory coupled to the controller; a pulse width modulation controller coupled to the controller and to a driver, the driver being coupled to a load; a loop filter coupled to the driver, an audio digital-to-analog converter (DAC) and an assist DAC, wherein the controller is further configured to control the assist DAC.

[0007] According to at least one aspect of the present disclosure, a system is provided having a pulse width modulation controller coupled to a driver, the driver configured to provide pulses, based on control signals from the pulse width modulation controller, to a first rail and a second rail; a load coupled between the first rail and the second rail; a first rail impedance incorporated into the first rail; a second rail impedance incorporated into the second rail; a first digital-to-analog converter (DAC) with a first output coupled to the first rail and a second output coupled to the second rail; a first integrator with a first input coupled to the first rail and a second input coupled to the second rail, and a first output coupled to a third rail, and a second output coupled to a fourth rail; a second integrator with a first input coupled to the third rail and a second input coupled to the fourth rail; an assist DAC with a first output coupled to the third rail and a second output coupled to the fourth rail; and a controller configured to control the assist DAC.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

[0009] FIG. 1 illustrates a loop filter system according to an example;

[0010] FIG. 2 illustrates a loop filter system according to an example;

[0011] FIG. 3A illustrates a process for finding an off resistance according to an example;

[0012] FIG. 3B illustrate a process for finding an off resistance according to an example;

[0013] FIG. 4A illustrates a process for finding an on resistance according to an example;

[0014] FIG. 4B illustrates a process for finding an on resistance according to an example;

[0015] FIG. 4C illustrates a process for finding an on resistance according to an example;

[0016] FIG. 5A illustrates a process for determining an audio drive according to an example;

[0017] FIG. 5B illustrates a process for determining an audio drive according to an example;

[0018] FIG. 6 illustrates a chart showing a relationship between the on resistance and off resistance

according to an example; and

[0019] FIG. 7 illustrates a process for managing a circuit according to an example.

DETAILED DESCRIPTION

[0020] When an amplifier or load experiences a low load impedance (or, more generally, a load impedance that deviates by a threshold amount from an expected value), this may indicate a fault condition (such as a short circuit, overvoltage and/or current condition, and so forth). Low load impedance can cause the amplifier or load to perform poorly or even be damaged. For example, in audio applications, low load impedance may distort or reduce the quality of an audio signal (for example, a signal provided to the ear of a listener). To protect the amplifier or load, and to prevent poor operation of the components of the circuit, examples disclosed herein may shut down the amplifier. The sooner the amplifier is disabled (relative to the fault condition occurring), the less the chance that undesirable conditions, such as damage or poor operation, may occur.

[0021] Elements of this disclosure relate generally to rapidly identifying low impedance conditions and disabling affected amplifiers, including identifying said conditions across a wide range of levels (frequencies, phases, amplitudes, channels, and so forth).

[0022] Some existing methods may identify a voltage drop across a switching device to detect a low load impedance, with a large drop indicating a low impedance. These methods require extra analog circuitry to measure the voltage drop, and must measure the precise timing during which a pulse is occurring. In multirail systems, the extra circuitry (for measuring voltage drop and timing of the pulse) may need to be replicated for each voltage rail. Furthermore, these methods depend on peak current, which means that accuracy depends on pulse width, and so precise load value determination is difficult.

[0023] By contrast, methods and systems disclosed herein do not require extra circuitry, have more robust timing requirements, and are more accurate.

[0024] Aspects of the present disclosure involve the use of analog-to-digital converters (ADCs) and digital-to-analog converters (DACs), and digital computation to overcome the limitations of the above-described methods and systems. Consider a system that may operate in closed-loop mode as well as open-loop mode and may even operate in other modes. The system may include one or more drivers being used to drive an output (such as an audio output). The on-resistance (“ R_{on} ”) and off-resistance (“ R_{off} ”) of the driver may be estimated from the drive signals used to keep the system in a closed-loop state. From this, the ratio of R_{on} and R_{off} to the resistance of load (e.g., the “output resistance” or “load impedance”) may be determined. If the load resistance falls outside a given range, the system may determine that a short or fault condition has occurred and take remedial action, such as disabling a portion or all of the system.

[0025] In some examples, an analog-to-digital converter (ADC) is used to digitize the output of a loop filter. The digitized output of the loop filter drives pulse-width modulation pulses (PWM pulses) as well as an output audio signal. As a result, digital logic determines the desired pulse width, which is precisely known. Given that the supply voltage is also known, deviations in the pulse width from the desired pulse width may be estimated with precision. As a result, the volt-seconds for each pulse prior to the effect of driver output resistance may be known. Furthermore, if the system is operating in a closed-loop mode, the voltage and/or current on the load may be known. The ratio of these two voltages results from a resistive voltage divider formed by the driver resistance and the load resistance. This ratio may be accurately determined, and may be adjusted for external conditions, such as temperature. Thus, the load impedance may be calculated with the existing circuitry and some digital processing. As the loop filter has high bandwidth and gain, determining the load impedance is fast and accurate over a wide range of signal levels.

[0026] FIG. 1 illustrates a block diagram of a loop-filter system **100** (“system **100**”) configured to detect shorts or other fault conditions according to an example. The system **100** includes a driver **102**, a loop filter **104**, a pulse-width modulation controller **106** (“PVMC **106**”), a controller **108**, an assist DAC **110**, memory and/or storage **112** (“storage **112**”), one or more loads **114** (“load **114**”),

and an audio DAC **116** (“AUDAC **116**”).

[0027] The driver **102** is coupled to load **114**, loop filter **104**, and PWM **106**. The loop filter **104** is coupled to at least the driver **102** and PWM **106**, the assist DAC **110**, and may be coupled to the load **114**. The PWM **106** is coupled to the driver **102**, loop filter **104**, and controller **108**. The controller **108** is coupled to the PWM **106**, storage **112**, and assist DAC **110**. The assist DAC **110** is coupled to the controller **108** and loop filter **104**. The AUDAC **116** is coupled to the loop filter **104**.

[0028] The driver **102** can provide an output a voltage or current to the load **114** and loop filter **104**. The driver **102** may provide a DC or AC output, may provide the output in pulses (such as square or triangular pulses) having a duration and voltage, and so forth. In some examples, the driver **102** may be an H-bridge or similar device.

[0029] The loop filter **104** may process the output from the driver **102**. The loop filter **104** may, for example, attenuate or amplify the output or portions of the output (e.g., portions corresponding to given frequencies, phases, amplitudes, and so forth). The loop filter **104** provides an output to the PWM **106**. The output from the loop filter **104** may be a processed version of the output from the driver **102** (e.g., the output that is also provided to the load **114**).

[0030] The PWM **106** may control the driver **102**. The PWM **106** may generate a signal indicating when the driver **102** should provide an output and/or may generate a signal indicating when the driver **102** should stop providing an output. In some examples, the PWM **106** may provide a signal to the driver **102** indicating that the driver **102** should provide an output for a specific duration of time. The PWM **106** may receive a processed output from the loop filter **104**, and may use the processed output from the loop filter **104** to determine how the driver **102** will operate. For example, the PWM **106** may, based on feedback from the loop filter **104**, adjust the signals the PWM **106** provides to the driver **102**, thus causing the driver **102** to provide outputs with different voltage and/or currents, durations, or other characteristics. By controlling the driver **102** to change its output, the feedback (e.g., processed output) from the loop filter **104** may change, and may cause the PWM **106** to further adjust the behavior of the driver **102**. In this way, the PWM **106** can control the driver **102**, based on the feedback from the loop filter **104**, to reach and/or maintain a desired state for the system **100** as a whole.

[0031] The PWM **106** can also provide information to the controller **108**. For example, the PWM **106** may provide information indicative of or related to the way the PWM **106** is adjusting the behavior of the driver **102**.

[0032] The storage **112** may be RAM, ROM, one time programmable memory (“OTP memory”), registers, and so forth. In some examples, the storage **112** may contain values determined during testing, manufacture, production, or during calibration. The storage **112** may contain predetermined values corresponding to the trim values for the assist DAC **110** current scaling, the resistance (or, more generally, the impedance) of the driver **102**, and impedance values corresponding to the impedance of the load under various conditions. The storage **112** may also store other values calculated by the system **100**.

[0033] The assist DAC **110** may provide an output to the loop filter **104**. The assist DAC **110** may convert a digital input to an analog voltage and/or current. The analog voltage and/or current may be provided to the loop filter **104**, while the digital input may be received from the controller **108** or another driver that provides a signal to the assist DAC **110**, or from the driver **102** (after being converted to digital form by an ADC).

[0034] The load **114** may be any type of load, device, or circuit, including audio systems.

[0035] The AUDAC **116** is an audio DAC and may provide a voltage or current to the loop filter **104**.

[0036] The controller **108** controls at least the assist DAC **110**. The controller **108** may receive a signal from the PWM **106** that is indicative of how the PWM **106** will be or is controlling the driver **102**. The controller **108** may also access the storage **112** to retrieve values stored in the

storage **112**. During operation, the controller **108** may calculate various values and base its control of the assist DAC **110** on the calculated values. For example, during operation the controller **108** may calculate Ron and Roff for each rail of the driver **102** (that is, if driver **102** is an H-bridge topology with multiple rails for voltage and ground, the controller **108** may calculate Ron and Roff for each of those respective rails). The calculation of Ron and Roff for the rails of the driver **102** may be performed using drive signals being used to keep the loop closed. These drive signals may be received from the PWM **106** and may be indicative of how the PWM **106** is controlling the driver **102**. If the values of Ron and/or Roff deviate from a target value by a threshold amount, the controller **108** may disable the loop-filter system **100** or portions of the system.

[0037] FIG. 2 illustrates a portion of a loop-filter system **200** ("system **200**") according to an example. The system **200** includes a driver **202**, a driver controller **204**, a load **206**, a loop filter **207**, a first impedance **208**, a second impedance **210**, a first integrator **212**, a second integrator **214**, a differential and common mode DAC **216** ("first DAC **216**"), an assist DAC **218** ("second DAC **218**"), and a controller **220**.

[0038] The driver controller **204** is coupled to the driver **202**. The driver **202** is coupled at a positive rail to the load **206** and first impedance **208**, and at a negative rail to the load **206** and second impedance **210**. The load is coupled at a first connection to the positive rail of the driver **202** and to the first impedance **208**, and at a second connection to the negative rail of the driver **202** and to the second impedance **210**. The first impedance **208** is coupled at a first connection to the positive rail of the driver **202** and to the load **206**, and at a second connection to the first integrator and to the first DAC **216**. The second impedance **208** is coupled at a first connection to the negative rail of the driver **202** and to the load **206**, and at a second connection to the first integrator **212** and to the first DAC **216**. The first DAC **216** is coupled at a first connection to the first impedance **208** and the first integrator **212**, and at a second connection to the second impedance **210** and the first integrator **212**. The first integrator **212** is coupled at a first connection to the first impedance **208** and a first output of the first DAC **216**, at a second connection to the second impedance **210** and a second output of the first DAC **216**, at a third connection to the second integrator **214** and to a first output of the second DAC **218**, and at a fourth connection to the second integrator **214** and a second output of the second DAC **218**. The second integrator **214** is coupled at a first connection to the first integrator **212** and the first output of the second DAC **218**, and at a second connection to the first integrator **212** and the second output of the second DAC **218**. The second DAC **218** is coupled at a first connection to the first integrator **212** and second integrator **214**, and at a second connection to a different connection of the first integrator **212** and second integrator **214**. The controller **220** is coupled to the second DAC **218**.

[0039] In general, there are two conducting paths from the driver **202** to the loop filter **207**. A first conducting path corresponds to the positive rail of the driver **202**, the first impedance **208**, and positive inputs and outputs of the first integrator **212** and second integrator **214**. A second conducting path corresponds to the negative rail of the driver **202**, the second impedance **210**, and negative inputs and outputs of the first integrator **212** and second integrator **214**. The first DAC **216** is coupled to the first conducting path at one of its outputs, and to the second conducting path at another of its outputs. Likewise, the second DAC **218** is coupled to the first conducting path at one of its outputs, and to the second conducting path at another of its outputs.

[0040] The driver **202** has at least one positive rail and at least one negative rail. The driver **202** provides an output voltage or current to at least one of the rails, and may provide output voltages or currents to any number of the positive and negative rails at any time. The driver **202** is controlled by the driver controller **204**. The driver controller **204** may be a pulse-width-modulation controller.

[0041] The load **206** may use current or voltages provided by the driver **202** to operate.

[0042] The first impedance **208** may have a resistance and/or reactance. The second impedance **210** may have a resistance and/or reactance as well. In some examples, the first impedance **208** and second impedance **210** have equal resistances and/or reactances, and in some examples they have

different resistance and/or reactances.

[0043] The first integrator **212** and second integrator **214** may be types of amplifiers that amplify or attenuate input voltages and/or currents. The first integrator **212** and second integrator **214** may have differential inputs and may provide differential outputs.

[0044] The first DAC **216** can provide a current and/or voltage to the first conducting path at a point between the first impedance **208** and the first integrator **212**, and to the second conducting path at a point between the second impedance **208** and the first integrator **212**. The first DAC **216** may provide a differential current and/or a common mode current.

[0045] The second DAC **218** can provide a current and/or voltage to the first conducting path at a point between the first integrator **212** and second integrator **214**, and to the second conducting path at a point between the first integrator **212** and the second integrator **214**.

[0046] In some examples, where the driver **202** has multiple positive rails and/or negative rails, the first DAC **216** and/or second DAC **218** may include connections to conducting paths corresponding to each of those rails.

[0047] The controller **220** may determine R_{on} and R_{off} for each of the positive and negative rails of the driver **202**, and may determine if a fault condition exists. In some examples, the controller **220** may determine R_{off} based on one or more of the differential current, common mode current, common mode input to the first integrator **212**, voltage rail values, and/or predetermined values. In some examples, the controller **220** may determine R_{on} based on one or more of various drive values (e.g., voltage and/or currents or currents provided by the driver **202**), a 100% duty cycle, an actual duty cycle, and an ideal duty cycle.

[0048] FIG. 3A illustrates a process **300** for determining R_{off} according to an example. Reference will be made to FIG. 2 in discussion of the process **300**.

[0049] At act **302**, the controller **220** determines the differential current provided by the first DAC **216**. The controller **220** may determine the differential current directly (e.g., using a current sensor), or may receive a value from the first DAC **216** or a controller of the first DAC **216** which indicates the value of the differential current. The process **300** may then continue to act **304**.

[0050] At act **304**, the controller **220** determines the common mode current provided by the first DAC **216**. The controller **220** may determine the common mode current directly (e.g., using a current sensor), or may receive a value from the first DAC **216** or a controller of the first DAC **216** which indicates the value of the common mode current. The process **300** may then continue to act **306**.

[0051] At act **306**, the controller **220** determines the differential input to an integrator. In some examples, the differential input may be related to the voltage and/or current provided on the second conducting path to the first integrator **212** (e.g., the voltage and/or current at node B in FIG. 2). In some examples, the controller **220** may determine the differential input as the value of the voltage and/or current at node B divided by the impedance of the second impedance **210**. In some examples, the voltage and/or current at node A divided by the impedance of the first impedance **208** may be used. Once the differential input has been determined, the process **300** may proceed to act **308**.

[0052] At act **308**, the controller **220** may determine the voltage (V_{svm}) used to control the PWM cycle of driver **202**. The process **300** may then proceed to act **310**.

[0053] At act **310**, the controller **220** may determine a value, referred to as “X,” which may be used to determine whether and how to change R_{off} . In some examples, X may be determined based on the differential input, the common mode current, and/or V_{svm} . One example of how to determine X is provided in FIG. 3B. The process **300** may then continue to act **312**.

[0054] At act **312**, the controller **220** may determine a value, referred to as “Y,” which may be used to determine whether and how to change R_{off} . In some examples, Y may be determined based on the differential current and a value (for example, the current value) of R_{off} . In some examples, Y may be determined as a product of the differential current and a value of R_{off} . The process **300**

may then continue to act **314**.

[0055] At act **314**, the controller **220** determines if X is greater than Y. In some examples, the controller **220** may determine if X is equal to Y as well. If the controller **220** determines that X is greater than Y (**314 YES**), the process **300** may continue to act **316**. If the controller **220** determines that X is less than (or in some examples, less than or equal to) Y (**314 NO**), the process **300** may continue to act **318**.

[0056] At act **316**, the value of Roff may be incremented by a set amount equal to or greater than a set value, for example 1 or 0.5. In some examples, the controller **220** may control the incrementing of the value of Roff stored in memory (such as a register).

[0057] At act **318**, the value of Roff may be decremented by a set amount equal to or greater than 1. In some examples, the controller **220** may control the decrementing of the value of Roff stored in memory (such as a register).

[0058] FIG. **3B** illustrates a process **350** for determining the value of “X” discussed with respect to act **310** of FIG. **3A** according to an example.

[0059] At act **352**, the controller **220** adjusts the value of the common mode voltage (V_{cm}). In some examples, the controller **220** may multiply the value of V_{cm} by a constant, such as 1 or -1 , based on the operating mode of the loop filter system **200**. For example, in a positive current drive mode the controller **220** may multiply the value of V_{cm} by -1 , while in other operating modes the controller **220** may multiply the value of V_{cm} by 1. The process **350** may then continue to act **354**.

[0060] At act **354**, the controller **220** determines the sum of V_{cm} (as adjusted in act **352**) and the common mode current (I_{cm}) by subtracting the common mode current from V_{cm} . That is, the controller **220** may determine the value of $(V_{cm} - I_{cm})$. The process **350** may then continue to act **356**.

[0061] At act **356**, the controller **220** may determine if the loop filter system **200** is operating in a positive current drive mode or a different mode. If the controller **220** determines that the loop filter system **200** is operating in a positive current drive mode (**356 YES**), the process **350** may continue to act **360**. If the controller **220** determines that the loop filter system **200** is not operating in a positive current drive mode (**356 NO**), the process **350** may continue to act **358**.

[0062] At act **358**, the controller **220** finalizes the value of X. In some examples, X may equal the difference between V_{cm} and I_{cm} : that is, X may equal $(V_{cm} - I_{cm})$. The process may then end.

[0063] Returning to act **356**, as stated above, the process may continue to act **360** if the controller **220** determines that the loop filter system **200** is operating a positive current drive mode (**356 YES**).

[0064] At act **360**, the controller **220** may scale V_{svm} by a constant. The constant may be preset and/or stored in memory, and may be based on characteristics of the second DAC **218**. The scaling constant may be greater than or equal to zero. The process **350** may then continue to act **362**.

[0065] At act **362**, the scaled value of V_{svm} is summed together by the controller **220** with the sum of V_{cm} and I_{cm} . For example, the scaled value of V_{svm} may be added to the sum of the differential input and common mode current, as expressed by the equation: $V_{cm} - I_{cm} + V_{scaled}$, where V_{scaled} is the scaled value of V_{svm} . The process **350** may then continue to act **368**.

[0066] At act **368**, the controller **220** finalizes the value of X. In some examples, X may equal the difference between the differential input and the common mode current further summed together with the scaled value of V_{svm} : that is, X may equal $V_{cm} - I_{cm} + V_{scaled}$.

[0067] FIG. **4A** illustrates a process **400** for determining the value of R_{on} according to an example.

[0068] At act **402**, the controller **220** determines the audio drive. The controller may determine the audio drive based on R_{on} and/or Roff, the differential current, and the scaled value of V_{svm} . In some examples, the value of the audio drive may depend on the operating mode of the system. A more detailed discussion of one method of how to calculate audio drive is presented with respect to FIG. **5A**. The process **400** may then continue to act **404**.

[0069] At act **404**, the controller **220** determines the pulse drive. The controller may determine the

pulse drive based on Ron and/or Roff, the differential current, and the scaled value of Vsvm. In some examples, the value of the pulse drive may depend on the operating mode of the system. A more detailed discussion of one method of how to calculate pulse drive is presented with respect to FIG. 5B. The process **400** may then continue to act **406**.

[0070] At act **406**, the controller **220** determines the amount of energy provided by the driver **202** during a 100% on duty cycle. The controller may determine the amount of energy provided during a 100% duty cycle by determining the voltage and/or current and/or charge delivered, and/or based on predetermined constants and time, and so forth. The process **400** may then continue to act **408**.

[0071] At act **408**, the controller **220** determines the nonoverlap interval for the driver **202**. That is, the driver **202** may have a finite turn on and turn off time, and thus may not reach full power instantaneously. For example, the voltage and/or current provided by the driver **202** may increase over a, usually short, period of time. The nonoverlap interval is an indication of the amount of time it takes for the driver **202** to turn on and off. The nonoverlap interval may be expressed as the percentage of time (of a 100% duty cycle, for example) that it takes the driver **202** to turn on, or it may be expressed a proportion of the time the driver is on compared to the total time or a proportion of the time the driver is off compared to the total time, or in any other desired way. Each of these expressions can indicate the amount of time the driver **202** is operating in a given mode (e.g., off, on, or ramping up/down from on to off). Once the nonoverlap interval is determined, the process **400** may continue to act **410**.

[0072] At act **410**, the controller **220** determines an ideal duty cycle. The ideal duty cycle may be the duty cycle for the driver **202** that would reflect a desired duty cycle assuming no turn-on or turn-off time (e.g., assuming that, once the driver **202** is activated, the driver **202** immediately reaches maximum output, and that once the driver **202** is deactivated, the driver **202** immediately ceases any output, without a finite ramp up or ramp down time). The process **400** may then continue to act **412**.

[0073] At act **412**, the controller **220** determines X. In some examples, the controller **220** may determine X based on the pulse drive, nonoverlap interval, 100% duty cycle, and/or ideal duty cycle. A process for determining X is discussed in greater detail with respect to FIG. 4B. Once the controller **220** determines X, the process **400** may continue to act **414**.

[0074] At act **414**, the controller determines Y. In some examples, the controller **220** may determine Y based on the 100% duty cycle, nonoverlap interval, ideal duty cycle, audio drive, and/or operating mode. A process for determining Y is discussed in greater detail with respect to FIG. 4C. Once the controller **220** determines Y, the process **400** may continue to act **416**.

[0075] At act **416**, the controller **220** determines if X is greater than Y. In some examples, the controller **220** may determine if X is equal to Y as well. If the controller **220** determines that X is greater than Y (**416 YES**), the process **300** may continue to act **418**. If the controller **220** determines that X is less than (or in some examples, less than or equal to) Y (**416 NO**), the process **300** may continue to act **420**.

[0076] At act **418**, the value of Ron may be incremented by a set amount equal to or greater than a set value, for example 1 or 0.5. In some examples, the controller **220** may control the increment of the value of Ron stored in memory (such as a register).

[0077] At act **420**, the value of Ron may be decremented by a set amount equal to or greater than a set value, for example 1 or 0.5. In some examples, the controller **220** may control the decrement of the value of Ron stored in memory (such as a register).

[0078] FIG. 4B illustrates a process **430** for determining a value of X (such as during act **412** of process **400** of FIG. 4A) according to an example.

[0079] At act **432**, the controller **220** may adjust the 100% duty cycle based on the nonoverlap interval. For example, the nonoverlap interval may be a value expressing a finite turn-on or turn-off time for the driver **202**. The nonoverlap interval may therefore, in some examples, be a value between 0 and 1, with 1 indicating that the driver **202** turns on instantaneously, and 0 indicating

that the driver **202** never turns on. The 100% duty cycle may be multiplied by the nonoverlap interval to produce an adjusted duty cycle. The process **430** may then continue to act **434**.

[0080] At act **434**, the controller **220** may sum together the adjusted duty cycle and the ideal duty cycle to produce the adjusted drive value. In some examples, the adjusted duty cycle is subtracted from the ideal duty cycle. The process **430** may then continue to act **436**.

[0081] At act **436**, the controller **220** may modify the adjusted drive value by the pulse drive. In some examples, the adjusted drive value and the pulse drive may be multiplied together. The process **430** may then continue to act **438**.

At act **438**, the controller **220** may output the value of X based on the previous acts **432-436**. In some examples, $X = (DC.sub.ideal - (DC.sub.100\%.Math.n)).Math.D.sub.pulse$, where DC.sub.100% is the 100% duty cycle, n is the nonoverlap interval, DC.sub.ideal is the ideal duty cycle, and D.sub.pulse is the pulse drive.

[0082] FIG. **4C** illustrates a process **450** for determining a value of Y (such as during act **414** of process **400** of FIG. **4A**) according to an example.

[0083] At act **452**, the controller **220** determines whether the system is operating in a voltage drive mode or a current drive mode. If the controller **220** determines that the system is operating in a voltage drive mode (**452 YES**), the process **450** may continue to act **454**. If the controller **220** determines that the system is not operating in a voltage drive mode (**452 NO**), the process **450** may continue to act **456**.

[0084] At act **454**, the controller **220** sums the 100% duty cycle with the adjusted drive value. The adjusted drive value may be the same as the adjusted drive value determined during act **434** of process **430** of FIG. **4B**. In some examples, the controller **220** may subtract the adjusted drive value from the 100% duty cycle to produce the adjusted 100% duty cycle. The process **450** may then continue to act **456**.

[0085] At act **456**, the controller **220** may modify the audio drive by either the 100% duty cycle, or the adjusted 100% duty cycle. For example, the controller **220** may multiply the audio drive by the 100% duty cycle in some modes of operation (such as non-voltage drive modes of operation), or may multiply the audio drive by the adjusted 100% duty cycle in some modes of operation (such as voltage drive modes of operation). The process **450** may then continue to act **458**.

[0086] At act **458**, the controller **220** may output the value of Y based on the previous acts **452-456**. In some examples, $Y = DC.sub.100\%.Math.AD$, where DC.sub.100% is the 100% duty cycle and AD is the audio drive. In some examples, $Y = ((DC.sub.100\% - Adj).Math.AD)$, where Adj is the adjusted drive value.

[0087] FIG. **5A** illustrates a process **500** for determining the audio drive according to an example.

[0088] At act **502**, the controller **220** determines the mode of operation of the system. In some examples, the controller **220** determines whether the system is in a voltage drive or current drive mode of operation. If the controller **220** determines that the system is in a voltage drive mode of operation (**502 YES**), the process **500** continue to act **510**. If the controller **220** determines the system is not in a voltage drive mode of operation (**502 NO**), the process **500** continues to act **504**.

[0089] At act **504**, the controller **220** sums Ron and Roff to produce a summed value using whatever value of Ron and Roff is currently available (e.g., the most recently determined values for Ron and Roff). In some examples, the controller **504** may add Roff to Ron. The process **500** then continues to act **506**.

[0090] At act **506**, the controller **220** increments the summed value. In some examples the controller **220** increments the summed value by a constant amount. In some examples, the constant amount is 1, and in other examples, the constant amount may be any value greater than or equal to zero. The process **500** may then continue to act **508**.

[0091] At act **508**, the controller **220** modifies the summed value based on the differential current. In some examples, the controller **220** multiplies the summed value by the differential current value. The process **500** may then continue to act **516**.

[0092] At act **510**, the controller **220** modifies Roff. In some examples, the controller **220** may modify Roff by multiplying Roff by a constant value. In some examples, the constant value is 2. In some examples, the constant value may be any value greater than or equal to zero. The process **500** may then continue to act **512**.

[0093] At act **512**, the controller **220** increments Roff. In some examples, the controller **220** may increment Roff by a constant value. In some examples, the constant value is 1. In some examples, the constant value may be any value greater than or equal to zero. The process **500** may then continue to act **514**.

[0094] At act **514**, the controller **220** modifies Roff based on the value of the differential current. In some examples, the controller **220** may multiply Roff and the value of the differential current together. The process **500** may then continue to act **516**.

[0095] At act **516**, the controller outputs the audio drive value. In some examples, the audio drive value may equal the value determined during act **508**. In some examples, the audio drive value may equal the value determined during act **514**. For example, the value determined during act **508** may be used when the system does not operate in a voltage drive mode, while the value determined during act **514** may be used when the system is operating in a voltage drive mode. For example, in a voltage drive mode the following equation may provide the audio drive value: $AD = (((R_{sub.off} \cdot Math.m) + n) \cdot Math.I_{sub.diff})$, where $R_{sub.off}$ is Roff, AD is the audio drive, m is a first constant, n is a second constant, and $I_{sub.diff}$ is the differential current. For example, in a mode other than a voltage drive mode the following equation may provide the audio drive value: $AD = (((R_{sub.on} + R_{sub.off}) + n) \cdot Math.I_{sub.diff})$, where $R_{sub.on}$ is Ron.

[0096] FIG. 5B illustrates a process **550** for determining the pulse drive according to an example.

[0097] At act **552**, the controller **220** determines whether the system is in a voltage drive mode of operation. If the controller **220** determines that the system is in a voltage drive mode of operation (**552 YES**), the process **550** may continue to act **556**. If the controller **220** determines that the system is not in a voltage drive mode of operation (**552 NO**), the process **550** may continue to act **554**.

[0098] At act **554**, the controller **220** determines the scaled Vsvm. The process **550** may then continue to act **564**.

[0099] At act **556**, the controller **220** may determine a sum of Ron and Roff to produce a summed value. In some examples, the controller **220** may add Ron and Roff together. The controller **550** may use a most recent value of Ron and a most recent value of Roff or whichever values of Ron and Roff are currently available. The process **550** may then continue to act **558**.

[0100] At act **558**, the controller **220** may increment the summed value by a constant amount. In some examples, the constant amount is 1. In some examples, the constant amount is any amount greater than or equal to zero. The process **550** may then continue to act **560**.

[0101] At act **560**, the controller **220** may modify the summed value based on the differential current. In some examples, the controller **220** may multiply the summed value and the differential current together to produce a product. The process **550** may then continue to act **562**.

[0102] At act **562**, the controller **220** may sum together the product and the scaled Vsvm. In some examples, the controller **220** may subtract the product from the scaled Vsvm. The process **550** may then continue to act **564**.

[0103] At act **564**, the controller **220** may provide the pulse drive. In some examples, the pulse drive value may be based on just the scaled Vsvm value (e.g., when the system is not in a voltage drive mode). For example, the following equation may express the value of the pulse drive: $PD = V_{sub.scaled}$, where PD is the pulse drive and $V_{sub.scaled}$ is scaled Vsvm. In some examples, the pulse drive may be based on Ron, Roff, scaled Vsvm, and the differential current. For example, the following equation may express the value of the pulse drive: $PD = V_{sub.scaled} - I_{sub.diff} \cdot Math.((R_{sub.on} + R_{sub.off}) + n)$, where $R_{sub.on}$ is Ron, $R_{sub.off}$ is Roff, n is a constant value, and $I_{sub.diff}$ is the differential current.

[0104] FIG. 6 illustrates a chart **600** showing the relationship between the calculated Ron or Roff value and the resistance of the load in Ohms, according to an example. The chart **600** includes a first axis **602**, a second axis **604**, a first trace **606**, a second trace **608**, a third trace **610**, and a fourth trace **612**.

[0105] The first axis **602** corresponds to the calculated value of either Ron or Roff (whichever is being used). The second axis **604** corresponds to the resistance of the load in Ohms. Note the second axis **604** is a logarithmic scale, where each tick along the second axis **604** represents an increase in the order of magnitude (thus, the first tick is 1, the second 10, the third 100, and the fourth is 1000).

[0106] The first trace **606** corresponds to a driver having a resistance of 2 Ohms. The second trace **608** corresponds to a driver having a resistance of 1.5 Ohms. The third trace **610** corresponds to a driver having a resistance of 1 Ohm. The fourth trace **612** corresponds to a driver having a resistance of 0.5 Ohms. However, these resistances are purely for explanatory purposes, and could be any value (or could be impedances having complex components).

[0107] The resistance of the driver may be stored in memory and may be determined prior to use (e.g., during testing of a particular device using the systems and methods discussed herein or using a driver such as driver **202**).

[0108] Consider an example where the load is 16 Ohms. For the third trace **610**, prior to use (e.g., during testing), the value of the third trace **610** may be approximately 0.105 at 16 Ohms of nominal load resistance. Suppose a threshold of 10 Ohms is used. At what value should the controller **220** consider a short to have occurred? In this example, a value for Ron or Roff (whichever is being used) should be 0.168. That is, Ron or Roff should remain above 0.168. This value may be determined by the equation:

$$[00001] T_{\text{fault}} = \text{OTP}_{\text{driver}} \cdot \text{Math.} \frac{T_{\text{Ohms}}}{\text{OTP}_{\text{Ohms}}} \quad (1)$$

[0109] In equation (1), above, T.sub.fault is the value above which Ron or Roff should remain. If Ron or Roff fall below T.sub.fault, this may indicate that a short or other fault condition has occurred. OTP.sub.driver indicates a value, stored in memory and determined prior to use (e.g., during testing) of Ron or Roff at a nominal load of k Ohms, where k may be chosen as any value and was chosen as 16 Ohms in the foregoing example. T.sub.ohms indicates the threshold in Ohms; in the foregoing example, it was chosen as 10 Ohms. OTP.sub.ohms

[0110] In general, since Ron and Roff represent a ratio of resistances, specifically the resistance of the driver to the resistance of the load, it is possible to calculate the level of resistance of the driver at a given moment in time by comparing Ron or Roff to the load in Ohms and working backwards to determine the driver resistance. If the driver resistance is too low (e.g., if Ron or Roff is too low when expressed as driver resistance over total resistance and/or load resistance), then this may indicate a short or other fault condition.

[0111] FIG. 7 illustrates a process **700** for managing a circuit using the techniques and systems described herein.

[0112] At act **702**, a controller determines a resistance, R, where R may be at least one of Ron or Roff. To determine Ron and/or Roff, the controller may implement physical hardware, physical hardware and software, or software to perform the techniques for determining Ron and Roff disclosed here. For example, the controller may implement the processes **300, 350, 400, 430, 450, 500, 550** of FIGS. 3A, 3B, 4A, 4B, 4C, 5A, and/or 5B described herein. The process **700** may then continue to act **704**.

[0113] At act **704**, the controller determines whether R is greater than or less than a threshold value. If R is greater than the threshold value, the process **700** may return to act **702** and determine R again at a future time (**704 YES**). Prior to returning to act **702**, the controller may perform other operations as well. If R is less than the threshold value, the process **700** may continue to act **706** (**704 NO**).

[0114] At act **706**, the controller may take corrective action. For example, the controller may determine that, since R is less than the threshold value, a fault has occurred. Accordingly, the controller may raise an alarm (e.g., physical, auditory, readable, visual, and so forth) to alert a user or another apparatus of the fault condition, or may attempt to protect the circuit by cutting power, reducing power use, temporary pausing operation of the circuit, and so forth.

[0115] Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

[0116] Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

[0117] References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for irreconcilable differences, the term usage in this document controls.

[0118] Various controllers, such as the controller **108**, may execute various operations discussed above. Using data stored in associated memory and/or storage, the controller **108** also executes one or more instructions stored on one or more non-transitory computer-readable media, which the controller **108** may include and/or be coupled to, that may result in manipulated data. In some examples, the controller **108** may include one or more processors or other types of controllers. In one example, the controller **108** is or includes at least one processor. In another example, the controller **108** performs at least a portion of the operations discussed above using an application-specific integrated circuit tailored to perform particular operations in addition to, or in lieu of, a general-purpose processor. As illustrated by these examples, examples in accordance with the present disclosure may perform the operations described herein using many specific combinations of hardware and software and the disclosure is not limited to any particular combination of hardware and software components. Examples of the disclosure may include a computer-program product configured to execute methods, processes, and/or operations discussed above. The computer-program product may be, or include, one or more controllers and/or processors configured to execute instructions to perform methods, processes, and/or operations discussed above.

[0119] Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of, and within the spirit and scope of, this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A system for detecting fault conditions, the system comprising: a driver having a first rail and a second rail, the driver configured to provide a first voltage to the first rail and a second voltage to the second rail; a loop filter coupled to the first rail at a first connection and to the second rail at a second connection; a first digital-to-analog converter (DAC) having a first DAC output coupled to the first connection and configured to provide a first DAC current to the first connection, and having a second DAC output coupled to the second connection and configured to provide a second DAC current to the second connection; and an assist DAC having a first assist output coupled to the loop filter and configured to provide a first assist current and having a second assist output coupled to the loop filter and configured to provide a second assist current.
2. The system of claim 1 wherein the system further comprises a controller and the controller is configured to adjust the output of the first DAC based on an on resistance and an off resistance.
3. The system of claim 2 wherein the controller is further configured to determine a value of the off resistance by: determining a first value based on a differential current and a stored value of the off resistance, the differential current being based on currents associated with the first rail and the second rail; and determining a second value based on a common mode voltage and a common mode current, the common mode voltage and the common mode current based on voltages and currents associated with the first rail and the second rail.
4. The system of claim 3 wherein the controller is configured to increase the value of the off resistance responsive to determining that the second value is greater than the first value.
5. The system of claim 3 wherein the controller is configured to decrease the value of the off resistance responsive to determining that the second value is less than the first value.
6. The system of claim 3 wherein the second value is further based on a scaled value of a control voltage used to control a pulse-width-modulation of the driver.
7. The system of claim 6 wherein the second value is equal to the common mode voltage less the common mode current and plus the scaled value of the control voltage.
8. The system of claim 3 wherein the second value is equal to the common mode voltage less the common mode current.
9. The system of claim 3 wherein the first value is equal to a product of the differential current and the stored value of the off resistance.
10. The system of claim 2 wherein the controller is configured to determine a value of the on resistance by: determining a first value based on a 100% duty cycle pulse width modulation and an audio drive value; and determining a second value based on the 100% duty cycle pulse width modulation, a non-overlap factor, an ideal duty cycle, and a pulse drive value.
11. The system of claim 10 wherein the on resistance is incremented responsive to determining that the second value is greater than the first value.
12. The system of claim 10 wherein the on resistance is decremented responsive to determining that the second value is less than the first value.
13. The system of claim 10 wherein the first value is further determined based on the non-overlap factor and the ideal duty cycle.
14. The system of claim 13 wherein the first value is product of the audio drive and first compositive value, wherein the first composite value is the 100% duty cycle pulse width modulation minus a second composite value, wherein the second composite value is a sum of the ideal duty cycle less a product of the 100% duty cycle pulse width modulation and the non-overlap factor.
15. The system of claim 10 wherein the first value of a product of the 100% duty cycle pulse width modulation and the audio drive.
16. The system of claim 10 wherein the second value is the pulse drive value times an adjusted drive value, wherein the adjusted drive value is the ideal duty cycle less the product of the 100% duty cycle pulse width modulation and the non-overlap factor.

- 17.** A method of detecting and addressing a fault condition in a circuit, comprising: determining a resistance based on at least one of an on resistance or an off resistance; comparing the resistance to a threshold value; determining that a fault has occurred responsive to determining that the resistance is less than the threshold value; and providing an alert signal indicating that the resistance is less than the threshold value.
- 18.** The method of claim 17 further comprising determining a present value of the off resistance by determining a differential current value provided by a first digital-to-analog converter (“first DAC”); determining a common mode current; determining a differential input to an integrator; determining a voltage used to control a pulse-width modulation cycle of a driver; determining a first value based on at least one of the differential input, the common mode current, or the common mode voltage; determining a second value based on at least one of the differential current and a predetermined off-resistance value; increasing the present value of the off resistance responsive to determining that the first value is greater than the second value; and decreasing the present value of the off resistance responsive to determining that the first value is not greater than the second value.
- 19.** The method of claim 18 wherein the first value is determined based on a difference between a common mode voltage and the common mode current and the second value is based on a sum of a first duty cycle and an adjusted drive value.
- 20.** The method of claim 17 wherein determining a present value of the on resistance by determining an audio drive value; determining an operating mode; determining a pulse drive value; determining a first duty cycle; determining a nonoverlap interval; determining an ideal duty cycle; determining a first value based on at least one of the pulse drive value, the nonoverlap interval, the first duty cycle, or the ideal duty cycle; determining a second value based on at least one of the audio drive value, the operating mode, the pulse drive value, the first duty cycle, the nonoverlap interval, the ideal duty cycle; increasing the present value of the on resistance responsive to determining that the first value is greater than the second value; and decreasing the present value of the on resistance responsive to determining that the first value is not greater than the second value.
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